

## General Guidelines

### MCA Program

The Master of Computer Applications (MCA) is a 3-year postgraduate program that aims to provide a solid foundation in computer science, technical expertise in software development, and practical skills in emerging technologies like cloud computing, machine learning, and big data. It emphasizes developing problem-solving abilities, fostering innovation, and preparing students to design secure, scalable IT solutions, ensuring industry readiness for dynamic careers in Information Technology. This program offers a balanced and comprehensive education in core computer science principles, specialized topics, and practical applications. Below is the area-wise credit distribution and the general structure of the program.

#### Academic Session

Each academic session is divided into two semesters of approximately eighteen weeks duration (with at-least seventy working days for classes in each semester):

1. Autumn Semester (July-November)
2. Winter Semester (January-April)

In addition, there may be a semester during the summer break, called a Summer Semester (May-June). The Senate approves schedule of academic activities for a session, inclusive of dates for registration, mid-semester and end-semester examinations, inter-semester breaks etc. The schedule is laid down in the Academic Calendar for the session.

#### Registration

At the beginning of each semester, until the completion of the program, a student must register for the semester and for the courses that he/she will study during the semester.

#### Eligibility for Course Registration

A student with no backlog courses (*i.e.* who has passed all the previous courses) will be eligible to register for all courses prescribed in the curriculum for the semester. A student who has backlog course(s) or is on academic probation may be recommended a different set of courses by the Dean of Academic Programs (Dean-AP).

#### Pre-requisite Courses

A student registering for a course must have successfully completed the prerequisite course(s), if any, for that particular course. For hard prerequisite, a minimum grade of DD is required.

#### Auditing of Courses

Auditing of courses allows students to gain exposure to additional subjects without increasing unduly their overall workload. Registration of courses for Audit is permitted from fifth semester onwards under the following conditions:

1. A student can audit a maximum of two courses during the entire program.
2. A student has to enter the courses to be audited in the Course Registration Form while registering for the semester. The word “Audit” would be specially mentioned in the remarks column of the student’s course registration form.
3. A student can register a course for audit provided the following two conditions are satisfied: (i) the course instructor permits and approves the registration, and (ii) the lecture, lab and tutorial time-table strictly permit.
4. An audit course will not be considered as an overload.
5. If the student’s performance is satisfactory, a grade of P (Pass) would be awarded. If the performance is not satisfactory, a grade of F (Fail) would be awarded.
6. An audit course will not be considered for the calculation of Semester Performance Index (SPI)/ Cumulative Performance Index (CPI). However, the course will be reflected in the Semester Grade Report and Transcript as an Audit Course provided a grade of P was obtained, otherwise the course will not appear in the Semester Grade Report and Transcript.

#### Course Load in Regular Semesters

A student is permitted to register for additional courses over the prescribed courses in the curriculum for a regular semester provided the total credits do not exceed 26. A student is permitted to under-load his/her prescribed academic load in a regular semester by dropping one or more courses provided the number of courses is at least 4 and the registered credits are not less than 12. However, after completion of his/her seventh regular semester, a student will be permitted to register for less than four courses.

#### Course Assessment

The assessment of students’ academic performance includes in-semester and end-semester examinations along with other continuous evaluation components. The various components of continuous assessment in a course may include home assignments, tutorial assignments, group assignments, quizzes, tests (open or closed book), viva-voce, mini projects, etc. Attendance in lectures/labs/tutorials may also be given due weightage in course assessment. The instructor may make attendance in lectures/tutorials/labs compulsory (80% or less) and after consulting the Dean-AP, award

‘F’ grade to students who do not achieve the prescribed level of attendance in that course.

The distribution of weightage, for the assessment of academic performance of students, through various modes listed above will be communicated by the course instructor at the beginning of the semester after taking due approval from the Director.

[*Note: Academic requirements such as projects and summer assignments, which are prescribed in the curriculum, are regarded as courses for the purpose of assessment.*]

### Letter Grade (10-point Scale)

For every course registered by a student, he/she is awarded a letter grade based on his/her combined performance in all the assessments. These letter grades are assigned points on a 10-point scale as described in the table below:

| Letter Grade | Points | Remark        |
|--------------|--------|---------------|
| AA           | 10     | Outstanding   |
| AB           | 9      | Excellent     |
| BB           | 8      | Very Good     |
| BC           | 7      | Good          |
| CC           | 6      | Average       |
| CD           | 5      | Below Average |
| DD           | 4      | Poor          |
| F            | 0      | Fail          |
| P            | -      | Pass          |
| I            | -      | Incomplete    |

A student passes the course if he/she gets any grade in the range of AA to DD, but fails if he/she gets the grade ‘F’. Certain courses are indicated as Pass/Fail courses, and in these courses a grade of ‘P’ or ‘F’ is awarded. ‘F’ grade may also be awarded in case of malpractice in examination/continuous evaluation process. Pass/Fail courses are not considered for calculation of SPI/CPI.

‘I’ grade will be awarded in a course if the overall performance of the student is satisfactory in the course, but the student either misses the end-semester examination due to illness, accident/death in the family or obtains such an approval from the Dean-AP under exceptional circumstances. A student who misses the end-semester examination must apply and his/her application must be supported (i) by proper medical certificate duly approved by the Medical Authority of the Institute in the case of illness, or (ii) by adequate evidence in the event of death in the family. An application not so supported will not be considered. Grade ‘I’ awarded for missing the end-semester examination will be converted into a performance grade

(depending on the overall performance of the student in the course) after taking an examination equivalent to the end-semester examination of that particular course. An ‘I’ grade must be converted into a performance grade by the specified date in the academic calendar for the next semester, otherwise it will be converted into ‘F’ grade.

### Repeating a Course

#### [A] As a Backlog Course

A student must repeat a course in which he/she has obtained an ‘F’ grade in a course taken for credit. Such a course is regarded as a backlog course and is subject to the regulations for registration. A backlog elective course can be replaced by another elective of the same category.

#### [B] For Grade Improvement

A student whose CPI is less than 5.0 is allowed to repeat a course in which a DD grade was obtained for the purpose of grade improvement in a regular semester only. The grade obtained in the repeated attempt(s) will be considered for the purpose of calculating the CPI for the semesters thereafter. However, the grade obtained in the first and subsequent attempt(s) will be shown in the Transcript.

### Performance Indices

#### [A] Semester Performance Index (SPI)

The performance of a student in a semester is indicated by the SPI. The SPI is the weighted average of the grade points obtained in all the courses registered by the student during the semester, calculated to two decimal places.

#### [B] Cumulative Performance Index (CPI)

An up-to-date assessment of the overall performance of a student from the time of entering the Institute is obtained by calculating the student’s CPI. The CPI is the weighted average of the grade points obtained in all the courses registered for credit by the student after entering the Institute. The CPI is also calculated to two decimal places.

#### Calculation of SPI and CPI

Let the course credits are  $U_1, U_2, \dots$  and the numeric values of the corresponding grade awarded in the courses are  $G_1, G_2, \dots$ , respectively, the SPI is given by

$$SPI = \frac{U_1 G_1 + U_2 G_2 + \dots}{U_1 + U_2 + \dots}$$

In the above computation, courses with P grades are ignored. Similarly, the CPI indicates the cumulative academic performance in all the courses taken including those taken in the current semester as

$$CPI = \frac{1}{Total\ Credits} \sum_{i=1}^8 (SPI \times Total\ credits\ of\ i^{th}\ semester)$$

### Final CPI and Class

For the purposes of computing the CPI at the end of the program, the student's CPI will be computed on the basis of the best CPI obtainable from the courses taken. The grade of B. Tech. Project (if graded as prescribed in the curriculum) should be included while computing the final CPI of the student. The Transcript will indicate *Distinction* if the student obtains a CPI of 9.00 or above and *First Class* if the student obtains a CPI of 6.50 or above but less than 9.00.

### Graduation Requirements

1. The student should have taken for credit and passed all the core and elective courses prescribed in the curriculum of the program.
2. The minimum CPI required for graduation is 5.0.
3. In case the student has completed the minimum credits and grade point requirements, but does not have the minimum CPI, then he/she must take additional elective courses or can take up core course(s) for grade improvement.
  - 3.1 If a student repeats a course for grade improvement, the grade is replaced only when the grade has improved.
  - 3.2 Grade for an elective course can be replaced by a grade of another elective course only if the minimum credit requirement is fulfilled.
4. The students should have passed all the core and Pass/ Fail courses prescribed in the curriculum, including all prescribed internships.
5. The student should have obtained a passing grade for the BTech project as prescribed in the curriculum.
6. The student should have registered for at least eight regular semesters (i.e. excluding summer semester, if any) as a regular student and should have paid all institute dues.
7. The student should have no case of indiscipline pending against him/ her.

### Award of Degree

The degree will be conferred on a student after he/she has fulfilled the graduation requirements stipulated in the curriculum (as approved by the senate).

### Award of Medals

#### [A] The Chairperson's Gold Medal

Student(s) excelling in academic performance with **CPI ( $\geq 7$ )** at the end of 6<sup>th</sup> semester without any backlog and overall contribution to the Institute activities comprising Campus Life, Social Services and Evidenced Leadership qualities would be

considered for the award of Chairperson's Gold Medal. In addition to the above criteria, the student must not have been awarded any deterrent by the Institute Disciplinary Action Committee throughout the program.

Based on an overall assessment, following due process of evaluation, recommendation of the Senate and approval of the Board of Governors (BoG) of the Institute, a candidate would be considered for the award of Chairperson's Gold Medal, in accordance with the Institute Rules.

#### [B] The Institute Medals

1. The Institute *Gold Medals* will be awarded to students who have secured first places in their respective programs.
2. The Institute *Silver Medals* will be awarded to the second place holders for each program.

### Poor Academic Performance

#### [A] Academic Probation

A student will be placed on Academic Probation for his/her second semester with written intimation if his/her SPI at the end of first semester is less than 4.5. In subsequent semesters, a student will be placed on Academic Probation with written intimation if his/her CPI in the previous semester is less than 5.0 or if his/her SPI is less than 4.5 in the previous semester. For every student placed on Academic Probation, the Dean-AP will prescribe a minimum SPI the student must attain in the semester. The minimum SPI so stipulated will be arrived at on the basis of the performance of the student in terms of her/his SPI/CPI as compared to the minimum requirements for graduation.

#### [B] Discontinued from the Institute on Account of Poor Academic Performance

If the performance of a student is poor so that he/she is not likely to benefit from continuing in the program any further, he/she would be required to leave the Institute. For this purpose an assessment of the student's academic performance will initially be made at the end of the second semester of his/her stay at the Institute and thereafter at the end of every subsequent semester. This assessment will be based on the CPI and SPI obtained by the student.

#### [C] Discontinued from the Institute on account of Poor Academic Performance at the end of the Second and Fourth Semester

A student whose CPI is less than 4.0 at the end of second or fourth semester shall be discontinued from the Institute. However, such a student may be allowed

to register for the available backlog courses offered in the summer semester, following his/her second or fourth semester. Such a student is permitted to register for a maximum of three of the available summer courses in which he/she is having F or DD grade. In case the student achieves the minimum CPI of 4.0 at the end of the relevant summer semester, he/she should be allowed to re-enter the program.

**Minimum and Maximum Period for Completion of MCA Program**

The minimum period to complete the program is three academic years. In any case, a student should fulfil the requirements for her/his degree within a maximum period of five academic years, failing which she/he will be required to leave the Institute. The period of five years excludes any semester in which the student has “withdrawn” status.

**Certificate of Academic Accomplishment**

A student who is unable to complete the degree requirements within the stipulated maximum period would be eligible to receive a “Certificate of Academic Accomplishment” by applying for it. The eligibility criteria and procedure for issue of the Certificate would be as laid down by the Institute senate from time to time.

---

## General Credit Distribution

| Sr. no. | Area                                                             | Credits/ Year  | I         | II        | III       | Proposed Credits |
|---------|------------------------------------------------------------------|----------------|-----------|-----------|-----------|------------------|
| 1       | Mathematics & Statistics                                         | 10-14          | 8         | 4         | 0         | 12               |
| 2       | Computer Science & Information Technology                        | 64-70          | 28        | 39        | 14        | 81               |
| 3       | Humanities and Social Sciences, Indian Knowledge System          | 10-14          | 5         | 3         | 0         | 8                |
| 4       | Soft Skills, Management Studies, Entrepreneurship and Innovation | 10-14          | 4         | 0         | 3         | 7                |
| 5       | Project / Internship                                             | 20-25          | 0         | 0         | 22        | 22               |
|         | <b>Total Credits</b>                                             | <b>128-134</b> | <b>45</b> | <b>43</b> | <b>39</b> | 127              |

| Credit convention                                  | Credits | Hours per week |
|----------------------------------------------------|---------|----------------|
| Lecture                                            | 1       | 1              |
| Tutorial                                           | 1       | 1              |
| Practicals/ Laboratory                             | 1       | 2              |
| Practicals/ Laboratory                             | 2       | 3              |
| Project / Research / Internship (for PI component) | 1       | 4              |

## Semester-wise Listing of Courses along with Credits

| Master of Computer Applications : Course List with Credit Structure |             |                                           |                                            |          |           |                   |
|---------------------------------------------------------------------|-------------|-------------------------------------------|--------------------------------------------|----------|-----------|-------------------|
| Semester                                                            | Course Code | Course Name                               | Credits ( Lecture - Tutorial - Practicals) |          |           | Total Credits (C) |
|                                                                     |             |                                           | L                                          | T        | P         |                   |
| I                                                                   | MS441       | Discrete Mathematics                      | 3                                          | 1        | 0         | 4                 |
|                                                                     | CS441       | Computer Programming and Problem Solving  | 3                                          | 0        | 3         | 5                 |
|                                                                     | CS443       | System Administration and Maintenance     | 3                                          | 0        | 2         | 4                 |
|                                                                     | EC441       | Digital Designs and Computer Architecture | 3                                          | 0        | 2         | 4                 |
|                                                                     | CS445       | Basics of Web Programming                 | 2                                          | 0        | 2         | 3                 |
|                                                                     | HS441       | Spoken and Written Communication          | 1                                          | 0        | 2         | 2                 |
|                                                                     | IK101       | Introduction to Yoga                      | 0                                          | 0        | 2         | 1                 |
|                                                                     |             | <b>Total Credits</b>                      | <b>15</b>                                  | <b>1</b> | <b>13</b> | <b>23</b>         |
| II                                                                  | MS442       | Linear Algebra                            | 3                                          | 1        | 0         | 4                 |
|                                                                     | CS442       | Data Structures                           | 3                                          | 0        | 2         | 4                 |
|                                                                     | CS444       | Object Oriented Programming               | 3                                          | 0        | 2         | 4                 |
|                                                                     | CS446       | Database Management Systems               | 3                                          | 0        | 2         | 4                 |
|                                                                     | CS448       | Operating Systems                         | 3                                          | 0        | 2         | 4                 |
|                                                                     | HS442       | Technical writing                         | 0                                          | 1        | 2         | 2                 |
|                                                                     |             | <b>Total Credits</b>                      | <b>15</b>                                  | <b>2</b> | <b>10</b> | <b>22</b>         |
| III                                                                 | MS541       | Probability & Statistics                  | 3                                          | 1        | 0         | 4                 |
|                                                                     | CS541       | Design & Analysis of Algorithms           | 3                                          | 1        | 2         | 5                 |
|                                                                     | CS543       | Web Services                              | 1                                          | 1        | 2         | 3                 |
|                                                                     | CS545       | Software Engineering                      | 2                                          | 1        | 2         | 4                 |
|                                                                     | CS547       | Computer Networks                         | 3                                          | 0        | 2         | 4                 |
|                                                                     | HS541       | Science, Technology and Society           | 3                                          | 0        | 0         | 3                 |
|                                                                     |             | <b>Total Credits</b>                      | <b>15</b>                                  | <b>4</b> | <b>8</b>  | <b>23</b>         |

|    |       |                                 |           |          |           |           |
|----|-------|---------------------------------|-----------|----------|-----------|-----------|
| IV | CS542 | Knowledge Management Systems    | 3         | 0        | 3         | 5         |
|    | CS544 | Machine Learning                | 3         | 0        | 2         | 4         |
|    | CS546 | Information Security            | 3         | 0        | 2         | 4         |
|    | CS548 | Elective – 1                    | 3         | 0        | 2         | 3         |
|    | CS552 | Elective – 2                    | 3         | 0        | 2         | 4         |
|    |       | <b>Total Credits</b>            | <b>15</b> | <b>0</b> | <b>12</b> | <b>20</b> |
| V  | CS641 | Artificial Intelligence         | 3         | 0        | 2         | 4         |
|    | CS643 | Distributed and Cloud Computing | 3         | 0        | 2         | 4         |
|    | CS645 | Elective – 4                    | 3         | 0        | 0         | 3         |
|    | CS647 | Elective – 5                    | 3         | 0        | 2         | 4         |
|    | CS649 | Elective – 6                    | 3         | 0        | 0         | 3         |
|    | CS691 | Internship                      | 0         | 0        | 6         | 3         |
|    |       | <b>Total Credits</b>            | <b>15</b> | <b>0</b> | <b>12</b> | <b>21</b> |
| VI | CS692 | Project                         | 0         | 0        | 36        | 18        |
|    |       | <b>Total Credits</b>            | <b>0</b>  | <b>0</b> | <b>36</b> | <b>18</b> |
|    |       |                                 |           |          |           |           |

## Semester-1 Course Content

| Course Code | Course Name          | L – T – P : C |
|-------------|----------------------|---------------|
| MS441       | Discrete Mathematics | 3 – 1 – 0 : 4 |

**Objective:** To develop logical thinking and its application to computer science and understand the basic principles of sets, functions and counting. The subject enhances one's ability to reason and ability to present a coherent and mathematically accurate argument.

**Learning Outcomes:** On successful completion of this course, students should be able to:

- Write valid arguments using logical notation, Understand the basic principles of sets and relations, counting techniques and graph theory.
- Acquire ability to describe and validate computer programs in a formal mathematical manner.

**Prerequisite:** None

### Contents:

Module 1: Foundation: Propositional and predicate logic, logical equivalences, predicates and quantifiers, translation from language to logical expressions, nested quantifiers, set theory, set operations, set identities and functions, inverse and composition functions, graph of functions.

Module 2: Number Theory: Division operator, prime factorization, properties of prime numbers, prime number theorem, GCD and LCM, modular arithmetic and applications, sequences and summations.

Module 3: Counting: Permutation and combinations, pigeonhole principle, inclusion-exclusion principle, binomial theorem, Pascal identity and triangle.

Module 4: Mathematical Reasoning and Induction: Rules of inference, direct proof, proof by contradiction, proof by contrapositive, mathematical induction and second law of mathematical induction.

Module 5: Recursion: Introduction to recursive algorithm, recurrence relations, solving recurrence relations.

Module 6: Relations: Relations and their properties, applications and representations, equivalence relations, partial ordering, Hasse diagram.

Module 7: Graphs: Introduction and terminology, representation, isomorphism, connectivity, Warshall's algorithm, Euler and Hamilton path, shortest path.

### Text Book:

1. *Discrete Mathematics and its Application*, 7th Ed, K. Rosen, Tata McGraw Hill, **2011**.

### Reference Books:

1. *Discrete Mathematical Structure*, 4th Ed, B. Kolman, R.C. Busby and S. C. Ross, PHI, 2000.
2. *Discrete Mathematics*, Richard Johnsonbaugh, Prentice Hall, 2007.



3. *Mathematics: A Discrete Introduction*, 3rd Ed., Edward R. Scheinerman, Cengage Learning, 2006.

4. *Mathematical Structure for Computer Science*, 6th Ed, J. Gersting, Freeman, 2006.

| Course Code | Course Name                              | L – T – P : C |
|-------------|------------------------------------------|---------------|
| CS441       | Computer Programming and Problem Solving | 3 – 0 – 3 : 5 |

**Objectives:** Students will learn computer programming as a tool for effective problem solving, with a focus on developing structured programs using the C language. They will also gain hands-on experience in implementing basic data structures in C, enabling them to write efficient and organized code for various computational tasks.

**Learning Outcomes:** Upon successful completion, students will be able to

- Solve problems using programming concepts.
- Model real-life scenarios into computational problems.
- Handle input/output operations efficiently.
- Apply fundamental data structures.
- Understand the computing environment and its limitations.

**Prerequisite:** None

#### **Content:**

Module 1 - Introduction to computers, basic organization, computational thinking, and problem-solving. Programming basics: syntax, semantics, debugging, error types. Software types: system vs. application. Compilation vs. interpretation. C language basics: standards, data types (int, char, float, double), storage classes (auto, static, extern, register), operators, constants, identifiers, formatted I/O (printf, scanf), sizeof, typecasting, integer arithmetic

Module 2 - Control constructs: if/else, ternary, switch. Problems: max of numbers, GCD, swapping. Loops: for, while, do-while for problems like sum, max, deviation, sine computation. Arrays: 1D arrays, strings as char arrays, basic string functions (e.g., palindrome check).

Module 3 - Problem solving: sorting 0s/1s, partitioning, merging arrays, square roots. Recursion: factorial, Fibonacci (recursive & DP). Binary search. Nested loops and sorting. Bitwise operations for sets: membership, union, intersection. Macros and preprocessor.

Module 4 - Structures and typedef (e.g., complex numbers). Functions: arguments in main, call by value/reference, exponentiation (logarithmic). Multi-dimensional arrays (e.g., matrix operations). Command-line arguments, variable-length arguments.

Module 5 - Pointers: basics, arithmetic, void\*, arrays vs. pointers, malloc, linked list intro. Pointers to structures/functions, array of pointers. enum usage. File Handling: Text file operations—read, write, append, etc.

#### **Text Books:**

1. Let us C, Yashavant Kanetkar, Latest Edition, BPB Publication

2. The C Programming Language, Brian W. Kernighan and Dennis Ritchie, Latest Edition, Prentice Hall.

#### Reference Books:

1. Programming in ANSI C, E. Balagurusamy, Latest Edition, McGraw Hill
2. Education Solutions Limited, I. T. L. (2004). Introduction to Computer Science. India: Pearson Education.
3. How to Solve it by Computer, RG Dromey, PHI

#### Tentative List of Lab programs:

1. Basic C commands and First C program-printing hello world on the screen, programs related to basic arithmetic operations, swapping of numbers etc.
2. C Expressions: Programs involving concepts of C expressions like finding roots of quadratic equation, area of circle and simple interest calculation.
3. C operators: Programs requiring in-depth knowledge of various C operators (especially conditional operator, bitwise operators and sizeof operator).
4. Conditional statements: Programs with applications of c conditional statements: if, if else, nested if else, switch-case.
5. Arrays and Loops: C programs for performing various operations (finding maximum, second-maximum, minimum, reversing an array etc) on 1-D arrays and Applications of concepts of loops (leap year, palindrome, displaying prime numbers etc).
6. Functions and Recursions: Programs demonstrating use of functions (like adding N numbers, calculator etc) and Recursion (factorial, Fibonacci, GCD, binary search etc).
7. Strings, Pointers and Structures: Programs related to the following concepts: String manipulations, pointer to arrays, and pointer to functions and Structures
8. File Management: Programs related to file handling (Finding the number of characters, words and lines of given text file and File handling programs)

| Course Code | Course Name                           | L – T – P : C |
|-------------|---------------------------------------|---------------|
| EC441       | System Administration and Maintenance | 3 – 0 – 2 : 4 |

**Objectives:** The course introduces students to the roles and responsibilities of the system administrator in large organizations (e.g., data centers), and focuses on hands-on experience of IT infrastructure administration.

#### Learning Outcomes:

- To list the roles and responsibilities of a system administrator in organizations.
- To manage the resources (system, network, database, users, etc.) and security of an IT infrastructure.
- To use the UNIX/Linux utilities and scripting languages effectively.
- To design and manage policies governing IT infrastructure.

**Prerequisite:** CS301.

**Contents:**

**Module 1:** Roles and responsibility of the system administrator, Friction between UNIX and Linux, Linux distributions and examples, System- specific administration tools, Man pages and other authoritative documentations.

**Module 2: Scripting and the Shell:** Shell basics – Command editing, Pipes and redirection, Variables and quoting, Common filter commands, bash scripting, Regular expressions, Python scripting, Scripting best practices.

**Module 3: Bootstrapping the system:** Booting PCs, GRUB: The GR and Unified Boot loader, Booting to single-user mode, Working with startup scripts, init and its run levels, systemd, Rebooting and shutting down.

**Module 4: Access Control and Users:** Traditional UNIX access control, Modern access control - Role-based access control, SELinux: security-enhanced Linux, PAM: Pluggable Authentication Modules, Kerberos - thirdparty cryptographic authentication, Access control lists, Real-world access control – su and sudo, Pseudo-users other than root, The /etc/passwd file, The /etc/shadow file, The /etc/group file, Adding and removing users, Managing users with system-specific tools.

**Module 5: Controlling Processes:** Components of a process, The life cycle of a process, Signals, kill, Process states, nice and renice: influence scheduling priority, ps – monitor processes, Dynamic monitoring with top, prstat, and topas, The /proc filesystem, strace: trace system calls, Runaway processes.

**Module 6: Periodic Processes and Filesystem:** cron - schedule commands, The format of crontab files, Crontab management, Linux and Vixie-cron extensions, Common uses for cron. Pathnames, Filesystem mounting and unmounting, File types, File attributes - The permission bits, The setuid and setgid bits, The sticky bit.

**Module 7: Syslog and log files:** Syslog: the system event logger, Syslog architecture, Configuring syslogd, Syslog debugging, Alternatives to syslog, Linux kernel and boot-time logging, logrotate - manage log files, Condensing log files to useful information, Logging policies. Introduction and requirements for backups, Backup devices and media, Incremental backups, dump and restore, archiving programs – tar and dd, Commercial backup products.

**Module 8: Networking and Routing:** TCP/IP and its relationship to the Internet, Networking road map, Packet addressing, IP addresses, Routing, ARP, DHCP, Basic network configuration, Linux networking.

**Text Book:**

1. Evi Nemeth et al., *UNIX and Linux System Administration Handbook*, Fifth Edition, Publisher: Addison-Wesley Professional, 2017.

**Reference Books:**

1. Aileen Frisch, *Essential System Administration*, Third Edition, Publisher: O'Reilly Media, 2009.
2. Limoncelli et al., *The Practice of System and Network Administration*, Second Edition, Publisher: Addison-Wesley Professional, 2007

### Tentative List of Lab Exercises

Lab Exercise 1 involves installing and exploring different Linux distributions such as Ubuntu or CentOS. Students will use system documentation tools like `man`, `info`, and files in `/usr/share/doc` to understand command usage and identify system-specific administration utilities.

Lab Exercise 2 focuses on shell basics, where students will practice command-line operations including pipes, redirection, environment variables, quoting, and filters like `grep`, `awk`, and `sed`, followed by writing simple bash scripts for file management tasks.

Lab Exercise 3, students will develop more advanced scripts using both Bash and Python, implementing regular expressions for parsing system logs or configuration files and automating administrative tasks.

Lab Exercise 4 covers the system boot process, where students will analyze GRUB, explore `systemd`, investigate system messages using `dmesg` and `journalctl`, and practice modifying run levels and startup services.

Lab Exercise 5 introduces user and access control management by creating, modifying, and deleting users, editing `/etc/passwd`, `/etc/shadow`, and `/etc/group` files, and configuring permissions using tools like `sudo` and `PAM`.

Lab Exercise 6 focuses on process control, allowing students to monitor and manage processes using commands such as `ps`, `top`, `kill`, `nice`, `renice`, and `strace`, while also exploring the `/proc` filesystem.

Lab Exercise 7 involves scheduling tasks using `cron`, editing `crontab` files, and working with filesystem attributes by applying permissions, `setuid/setgid`, and sticky bits, as well as mounting and unmounting file systems.

Lab Exercise 8, students will configure and monitor logging with `syslog` or `rsyslog`, analyze logs in `/var/log`, create custom logging rules, and implement log rotation using `logrotate`.

Lab Exercise 9 covers system backups using tools like `tar`, `dd`, `dump`, and `restore`, where students will perform full and incremental backups and restore data from archives.

Lab Exercise 10 introduces networking fundamentals, where students will configure IP addresses, use diagnostic tools like `ping`, `netstat`, `traceroute`, `ip`, and `ss`, and demonstrate basic routing and DHCP configurations.

| Course Code | Course Name                              | L – T – P : C |
|-------------|------------------------------------------|---------------|
| EC441       | Digital Design and Computer Architecture | 3 – 0 – 2 : 4 |

**Objectives:** To study the fundamentals of digital logic design and computer architecture of digital computers (CPU, memory, I/O, software, pipelining and parallelism). Discussions will include digital logic, microprogramming and performance enhancement of processors. Such

knowledge leads to better understanding computer organization and architecture, can be used in the design and implementation of computer systems or as foundation for more advanced computer-related studies.

**Learning Outcomes:**

- The students will be able to understand the fundamentals of digital logic design and circuits.
- The students will be able to understand the design and implementation of 4. digital circuits.
- To provide hands-on experience of software and hardware.
- To familiarize the students with the processor interaction with other hardware and time management.
- To familiarize the students with pipeline architecture and instruction level parallelism.
- To familiarize the students with dynamic instruction scheduling and thread level parallelism.
- To familiarize the students with the memory system and its interaction with the processor.

**Prerequisite:** CS441**Content:**

Module 1 - Number systems and Boolean algebra: Different number systems, conversion between them, binary arithmetic, complement representation, floating-point representation, logic gates, Boolean algebra laws, De Morgan's theorem, Sum of Products (SOP) and Product of Sums (POS).

Module 2- Combinational and Sequential logic: logic minimization using K-map method, QuineMcCluskey method, multiplexers/demultiplexers, decoders/encoders, adders, ripple carry adder, carry-lookahead adder, multiplication - shift-and-add, Booth multiplier. Simple circuits with feedback, basic latches, clocks, Flip flops, shift register, synchronous and asynchronous counters, Finite State Machine (Moore/Mealy Machines).

Module 3 - Von Neumann: Functional units, ALU, data paths architecture, registers, instruction set architecture (ISA), addressing modes.

Module 4 - Pipelining and Parallelism: An overview of pipelining, throughput and speedup, pipelined data path and control, data dependency and hazard, control hazard and structural hazard, instruction level parallelism (ILP) concepts and challenges, basic compiler techniques for exposing ILP, ILP using dynamic scheduling, VLIW, superscalar architecture, overview of thread level parallelism.

Module 5 - Memory system and I/O: Principles of temporal and spatial locality; cache memories (address mapping, block size, replacement and store policy); virtual memory (page table, TLB); disk organization and data access from disk drive, programmed I/O, interrupt- driven I/O, Handshaking, DMA, interrupts.

**Text Books:**

1. Digital Design, Morris Mano, Prentice Hall, 2002.
2. Digital Fundamentals, 10 th Ed, Floyd T L, Prentice Hall, 2009.
3. Computer Architecture: A Quantitative Approach, 5th Edition By John L Hennessy & David A Patterson.

**Reference Books:**

1. Digital electronics: principles, devices and applications, Albert Paul Malvino and Jerald A. Brown, John Wiley & Sons, 2007.
2. Computer system architecture, Morris Mano, revised third edition, Pearson.
3. Computer Organization and Architecture, 8th Ed, Stallings, Pearson, 2010.

**Tentative List of Lab Experiments:**

1. Design and implementation of data path architecture and program counter.
2. Design and implementation of arithmetic logic unit (ALU).
3. Design and implementation of ripple carry adder and carry-look-ahead adder.
4. Design and implementation of Booth's multiplier and combinational multipliers.
5. To understand the basic principles of pipeline design for single instruction, including the problems of data and branch hazards.
6. To understand pipeline design for multiple instructions, including the problems of data and branch hazards.
7. To understand the multiple instructions (beq, lw and st) working using pipelined processors.
8. To understand the arithmetic instruction (add, mult, div, sub) working using instruction level parallelism.
9. To understand the implementation of instruction level parallelism using ScoreBoard algorithm.
10. To understand the implementation of instruction level parallelism using the Tomasulo algorithm.

**Text Books:**

1. Digital Fundamentals, 10th Ed, Floyd T L, Prentice Hall, 2009.
2. Computer Architecture: A Quantitative Approach, 5th Edition By John L Hennessy; David A Patterson.

**Reference Books:**

1. Computer system architecture, Morris Mano, revised third edition, Pearson.
2. Computer Organization and Architecture, 8th Ed, Stallings, Pearson, 2010.

| Course Code | Course Name               | L – T – P : C |
|-------------|---------------------------|---------------|
| CS447       | Basics of Web Programming | 2 – 0 – 2 : 3 |

**Objectives:** This course introduces fundamental concepts of web development, covering HTML, CSS, and JavaScript for creating interactive web pages. It explores front-end and back-end technologies, including basic server-side scripting and databases. The course emphasizes hands-on practice in building responsive and dynamic websites.

**Learning Outcomes:** On successful completion of this course, students should be able to:

- Gain knowledge of fundamental web technologies, including HTML, CSS, and JavaScript, for creating structured and styled web pages.
- Learn to enhance user experience using JavaScript for dynamic and interactive web content.
- Apply CSS frameworks and techniques to build mobile-friendly and visually appealing web pages.
- Understand the basics of server-side scripting and database connectivity for dynamic web applications.

**Prerequisite:** None

**Content:**

Module 1 - Introduction to Web Programming: Overview of the web and how it works, client-side vs. server-side programming, introduction to web browsers and developer tools

Module 2 - HTML – Structuring Web Pages: HTML syntax and document structure, HTML elements: headings, paragraphs, lists, tables, and forms, hyperlinks, images, and multimedia integration, semantic HTML and best practices

Module 3 - CSS – Styling Web Pages: Introduction to CSS and its role in web design, CSS selectors, properties, and units, box model, flexbox, and grid layout, responsive design with media queries

Module 4 – JavaScript: Basics of JavaScript: variables, data types, and operators, functions, loops, and conditionals, DOM manipulation and event handling

Module 5 - Web Forms and User Input: Creating and styling forms, form validation using HTML5 and JavaScript, handling user input and feedback

Module 6 - Introduction to Frontend Frameworks: Basics of Bootstrap or Tailwind CSS for faster development, introduction to JavaScript libraries (e.g., React)

Module 7 - Basics of Backend Development: Understanding how the web server works, introduction to server-side programming (e.g., Node.js), basics of databases and SQL, sending and receiving data using HTTP requests

Module 8 - Web Hosting and Deployment: Introduction to web hosting services, deploying websites using GitHub Pages, Netlify, or FTP, basics of domain names and web hosting.

**Text Books:**

1. HTML and CSS: Design and Build Websites, Jon Duckett, Wiley (2011).
2. Learning Web Design, Jennifer Robbins, O'Reilly (2018).

**Reference Books:**

1. JavaScript: The Definitive Guide, David Flanagan, O'Reilly (2020).
2. Web Development and Design Foundations with HTML5, Terry Felke-Morris, Pearson (2018).

**Tentative List of Lab Exercises:**

Lab Program 1 introduces students to how the web works by exploring the difference between client-side and server-side programming and using browser developer tools for inspection.

Lab Program 2 focuses on writing structured HTML documents with elements such as headings, paragraphs, lists, tables, images, and forms, along with the use of semantic tags.

Lab Program 3 introduces CSS by teaching students how to style HTML pages using various selectors, properties, the box model, and layout techniques like flexbox and grid.

Lab Program 4 covers the basics of JavaScript, where students will write scripts using variables, functions, loops, and conditions, and manipulate the DOM and handle user events.

Lab Program 5 emphasizes creating interactive web forms with HTML5 and JavaScript validation and handling user inputs dynamically.

Lab Program 6 allows students to design responsive and modern UIs using frontend frameworks like Bootstrap or Tailwind CSS, along with a basic introduction to React components.

Lab Program 7 covers the fundamentals of backend development by introducing Node.js for server-side logic, writing basic HTTP servers, and interacting with databases using SQL queries.

Lab Program 8 guides students through the process of deploying static websites using GitHub Pages and Netlify, and introduces basic hosting concepts and domain name management.

Lab Program 9 involves a mini-project where students create a complete responsive website using HTML, CSS, JavaScript, and optionally a frontend framework.

Lab Program 10 involves a full-stack mini-project where students integrate frontend forms with a Node.js backend and a database, demonstrating end-to-end functionality.

| Course Code | Course Name                      | L – T – P : C |
|-------------|----------------------------------|---------------|
| HS441       | Spoken and Written Communication | 1 – 0 – 2 : 2 |

**Course Objectives:** This course develops communicative competence in English by focusing on LSRW skills, enhancing professional and technical communication, improving interpersonal interactions, preparing students for interviews and presentations, and strengthening pronunciation through phonetics.

**Learning Outcomes:** By the end of this course, students will be able to:

- Apply principles of effective communication in writing and speech.
- Enhance active listening and comprehension in professional contexts.
- Write structured technical and business documents (reports, emails, résumés).
- Deliver impactful presentations using verbal and non-verbal techniques.
- Demonstrate soft skills like teamwork, leadership, and negotiation.
- Communicate effectively in cross-cultural and global professional settings.

**Prerequisite:** None

**Content:**

Module 1: Foundations of Communication- Principles of Effective Communication: Definition, types, and process of communication. Clarity, coherence, conciseness, and correctness in spoken and written communication. Grammar and Sentence Structure: Basics of grammar,



sentence construction, and vocabulary building. Common errors and strategies for improvement. Overcoming Communication Barriers: Addressing linguistic, cultural, and psychological barriers in professional and academic settings.

Module 2: Oral and Aural Communication- Active Listening and Comprehension: Strategies for effective listening, note-taking, and responding. Barriers to listening and methods to overcome them. Phonetics and Pronunciation: Basics of phonetics, phonemic transcription, and accent neutralization. Public Speaking and Group Discussions: Techniques for effective speech delivery, body language, and fluency. Structuring debates, discussions, and interview responses.

Module 3: Fundamentals of Written Communication - Writing Process and Mechanics: Planning, drafting, revising, and editing for clarity and precision. Technical and Business Writing: Reports, research abstracts, project proposals, business emails, and memos. Résumé and Cover Letter Writing: Best practices for job applications, including structuring and tailoring documents for professional opportunities.

Module 4: Presentation and Professional Communication - Effective Presentation Skills: Structuring presentations, voice modulation, and non-verbal communication. Using Visual Aids: PowerPoint presentations, infographics, and charts to enhance communication. Cross-Cultural and Workplace Communication: Adapting communication styles for diverse professional environments and ethical considerations in communication.

Module 5: Soft Skills and Career Development - Teamwork and Leadership Communication: Strategies for collaboration, conflict resolution, and decision-making in a professional setting. Negotiation and Persuasion Techniques: Effective communication in meetings, interviews, and workplace discussions. Industry Readiness: Mock interviews, group discussions, and real-world communication simulations to enhance career preparedness.

Module 6: Reading Comprehension and Analytical Skills - Focused Reading Practice: Exposure to extended samples of written English, including short stories, essays, and newspaper editorials. Identifying challenges related to vocabulary, idioms, and complex sentence structures. Reading Strategies: Skimming, scanning, intensive and extensive reading techniques for better comprehension. Testing Comprehension Skills: Providing students with multiple reading passages in advance for analysis. Assessing comprehension through unseen and seen passages in examinations. Critical Thinking and Interpretation: Developing the ability to analyze, interpret, and infer meaning from diverse texts to enhance academic and professional reading proficiency.

**Text Books:**

1. Dhanavel, S.P. English and Communication Skills for Students of Science and Engineering. Orient Blackswan Ltd., 2009.

2. Day, R.A. Scientific English: A Guide for Scientists and Other Professionals, 2nd ed. Universities Press, 2000.
3. Ladefoged, Peter. A Course in Phonetics, 5th ed. Harcourt, Brace, Jovanovich: Fort Worth, 2006.
4. Raman, Meenakshi & Sharma, Sangeeta. Technical Communication: Principles and Practice, 2nd ed. Oxford University Press, 2009.
5. Murphy, Herta, Herbert Hildebrandt, & Jane Thomas. Effective Business Communication. Tata McGraw Hill, 1997.

**Additional References:**

1. Guffey, Mary Ellen, and Dana Loewy. Essentials of Business Communication, 10th ed. Cengage Learning, 2016.
2. Anderson, Paul V. Technical Communication: A Reader-Centered Approach, 8th ed. Cengage Learning, 2013.
3. Swales, John M. & Feak, Christine B. Academic Writing for Graduate Students: Essential Tasks and Skills, 3rd ed. University of Michigan Press, 2012.
4. Carter, Ronald & McCarthy, Michael. Cambridge Grammar of English: A Comprehensive Guide. Cambridge University Press, 2006.
5. Crystal, David. The Cambridge Encyclopedia of the English Language. Cambridge University Press, 2018.

## Semester-2 Course Content

| Course Code | Course Name    | L – T – P : C |
|-------------|----------------|---------------|
| MS442       | Linear Algebra | 3 – 1 – 0 : 4 |

**Objective:** Students will be able to apply the concepts and methods described in the syllabus and solve problems using linear algebra.

**Learning Outcomes:** Upon successful completion of this course, student will be able to:

- Solve the system of linear equations and apply the theory of vector spaces to understand it.
- Solve large system of linear equations using computational techniques.
- Demonstrate competence with the basic ideas of linear algebra including concepts of linear systems, independence, theory of matrices, linear transformations, bases and dimension, eigenvalues, eigenvectors and diagonalization.

**Prerequisite:** None

### Content:

Module 1: Matrices and Linear systems: Matrix operations (addition, multiplication), Block-Partitioned Matrices and Block Operations, Elementary Row and Column Operations, Determinant of a Square Matrix, Properties of the Determinant Function, Cofactor Expansion, Rank of a Matrix, The System of Linear Equations:  $Ax=b$

Module 2: Canonical Factorizations: Eigenvalues and Eigenvectors, Companion Matrices and Characteristic Polynomial, Method of Danilevsky for Characteristic Polynomial, diagonalization-Matrices with a Full-Set of Eigenvectors, The Cayley-Hamilton Theorem, Triagonalization and Unitary Diagonalization of a Matrix, Schur's Lemma and the Spectral Theorem, QRDecomposition, QR-Algorithm for Hessenberg Matrices, Singular Value Decomposition.

Module 3: Vector Spaces: Vector Space over the set real numbers (Field), Linear Independence of Vectors, Bases in a Vector Space, Dimension of a Vector Space, Direct Sum IIIT Vadodara: UG Curriculum 9 Decomposition of a Vector Space, Linear Transformations, Change of Bases, Canonical forms, Rank of a Linear Transformation.

Module 4: Numerical methods: Iterative methods (Jacobi, GaussSeidel, Relaxation) for solving linear systems, computing of eigenvalues and eigenvectors.

### Text Book:

1. Introduction to Linear Algebra, Gilbert Strang, 5 th Ed, SIAM, 2016.

### Reference Books:

1. Linear Algebra, Kunze Ray, Hoffman Kenneth 2 nd Ed, Phi Learning, 2014.
2. Fundamentals of Matrix Computations, David S. Watkins, 3 rd ed, Wiley.

|             |                             |               |
|-------------|-----------------------------|---------------|
| Course Code | Course Name                 | L – T – P : C |
| CS444       | Object Oriented Programming | 3 – 0 – 2 : 4 |

**Objective:** This course introduces basic concepts of object-oriented programming principles, design techniques, and analysis tools.

**Learning Outcomes:** On completion of this course the student should be able to design and analyze real-world problems based on object-oriented principles.

**Prerequisite:** CS441.

**Content:**

Module 1: Introduction: Principles of OOD; programming Paradigms; benefits of OOD&P, applications of OOD; Classes and objects; access qualifiers; instance creation; constructors, parameterized constructors, overloaded constructors, constructors with default arguments, copy constructors, static class members, and static objects.

Module 2: Functions and Operators: Function prototyping, function components, passing parameters, inline functions, default arguments, overloaded function; array of objects, pointers to objects, dynamic allocation operators, dynamic objects; Operator overloading, overloading a unary and binary operator, overloading the operator using friend function, stream operator overloading, data conversion.

Module 3: Inheritance: Defining derived classes, single inheritance, protected data with private inheritance, multiple inheritance, multi-level inheritance, hierarchical inheritance, hybrid inheritance, multi-path inheritance, constructors in derived and base class, abstract classes, virtual function and dynamic polymorphism, virtual destructor.

Module 4: Exception Handling: Principle of exception handling, exception handling mechanism, multiple catch, nested try, re/throwing the exception.

**Text Books:**

1. Introduction to object-oriented programming, B. Timothy, Pearson, 2001.
2. Object Oriented Design and Patterns, C. Horstmann, John Wiley & Sons, 2005.

**Tentative List of Lab exercises**

Lab Program 1 introduces the fundamentals of Object-Oriented Design by implementing classes and objects, using access specifiers, and writing constructors including default, parameterized, and copy constructors.

Lab Program 2 focuses on creating static class members and static objects, and understanding their their scope and behavior in object-oriented programming.

Lab Program 3 emphasizes function prototyping, passing parameters, using inline functions, default arguments, and function overloading to enhance code reusability.

Lab Program 4 involves creating arrays of objects and dynamic memory allocation using pointers and the new/delete operators to manage object lifetimes.

Lab Program 5 demonstrates operator overloading by overloading unary and binary operators, including stream operators for input/output operations.

Lab Program 6 focuses on implementing friend functions for operator overloading and practicing type conversion between class and primitive data types.

Lab Program 7 covers inheritance by creating programs involving single, multilevel, and hierarchical inheritance and observing constructor invocation order.

Lab Program 8 explores advanced inheritance types such as multiple, hybrid, and multipath inheritance, and introduces abstract classes and virtual functions to demonstrate dynamic polymorphism.

Lab Program 9 includes writing programs that use virtual destructors and handle object destruction correctly in polymorphic scenarios.

Lab Program 10 introduces exception handling by writing programs using try-catch blocks, handling multiple and nested exceptions, and demonstrating rethrowing mechanisms.

|             |                 |               |
|-------------|-----------------|---------------|
| Course Code | Course Name     | L – T – P : C |
| CS446       | Data Structures | 3 – 0 – 2 : 4 |

**Objectives:** The course covers fundamental data structures, including arrays, linked lists, stacks, queues, heaps, binary trees, and graphs. Students will learn to implement generic data structures with key operations like insertion, deletion, and searching. Additionally, the course emphasizes algorithm design and analysis.

**Learning Outcomes:** On successful completion of this course, students should be able to:

- Design and implement various data structures.
- Analyze the strength and weakness of different data structures.
- Compare various sorting and searching algorithms.
- Design optimized solutions for computational problems using appropriate data structures.
- Implement data structures in real-world applications and enhance problem-solving skills.

**Prerequisite:** Basic programming language such as C or Python

#### **Content:**

Module 1 - Introduction: Representation of data on a computer, data types, abstract data types, array and linked list representations, Big 'O' notion, introduction to classes and objects.

Module 2 - List data structures: Stacks, queues, variants implementation using arrays and linked lists.

Module 3 - Sorting and searching algorithms: Comparison and non-comparison-based sorting algorithms, linear and binary searching algorithms, best-case, worst-case and average-case analysis. Hashing techniques and collision resolution,

Module 4 - Trees: Heaps, Binary Trees, Binary search trees (BST), Balanced BSTs: Red Black trees, AVL Trees, 2,3,4-trees, B Trees.

Module 5 - Graphs: Representation using adjacency matrices and adjacency lists, Graph searching algorithms BFS and DFS.

#### Text Books:

1. Data Structures, Algorithms and Applications in C++, Sartaj Sahni, Universities Press, 2nd Edition (2009).
2. Data Structures and Algorithm in JAVA, Robert Lafre, Pearsons, 2nd Edition (2002).

#### Reference Books:

1. Introduction to Algorithms, 3th Ed, Cormen, Lieserson and Rivest, PHI, 2011.
2. Data Structures and Algorithms, Aho, Hopcroft and Ullman, Addison-Wesley, 1999.

---

#### Tentative List of Lab Programs

Lab Program 1: Introduces the representation of data using primitive and abstract data types. Students implement arrays and linked lists, and analyze operations using Big O notation.

Lab Program 2: Focuses on implementing stacks and queues using both arrays and linked lists, including common operations such as push, pop, enqueue, and dequeue.

Lab Program 3: Involves creating circular queues and double-ended queues (dequeues) using both static and dynamic memory allocation.

Lab Program 4: Covers various searching techniques including linear search and binary search, along with performance analysis in best, worst, and average cases.

Lab Program 5: Requires implementing comparison-based sorting algorithms such as bubble sort, insertion sort, selection sort, merge sort, and quicksort.

Lab Program 6: Introduces hashing techniques by implementing hash tables with different collision resolution strategies such as linear probing and chaining.

Lab Program 7: Focuses on implementing binary trees and binary search trees, including operations like insertion, deletion, and traversal (inorder, preorder, postorder).

Lab Program 8: Covers advanced tree structures like AVL trees and Red-Black trees, where students will implement balancing operations.

Lab Program 9: Involves implementing heap structures and performing heap sort, as well as exploring other tree types like B-trees or 2-3-4 trees.

Lab Program 10: Introduces graph representations using adjacency matrices and adjacency lists, and includes implementation of BFS and DFS algorithms.

---

| Course Code | Course Name                 | L – T – P : C |
|-------------|-----------------------------|---------------|
| HS446       | Database Management Systems | 3 – 0 – 2 : 4 |

**Objectives:** The course aims the students will be able to list and understand the basic concepts of a relational database system. They can analyze database requirements, determine the entities involved and relationships among them. The students are able to efficiently and effectively organize, maintain and retrieve information from a database system.

**Learning Outcomes:** On completion of this course the student should be able to:

- List the basic concepts of relational database model, relational algebra, entity-relationship model, and SQL.
- Convert the entity-relationship model to relational database and formulate SQL queries on it.
- Optimize database design through different processes.

**Prerequisites:** CS102

**Content:**

Module 1: Introduction and Conceptual Modeling: Databases and database users; database system concepts and architecture; data modeling using the entity-relationship (ER) model; enhanced entity relationship.

Module 2: Data Storage and Indexing: Introduction, record storage, and primary file organization index structures for files; single level indexing; multilevel indexing.

Module 3: Relational Model: The relational data model; relational database constraints; relational algebra; relational calculus; relational database design by ER and EER; relational mapping; SQL; the relational database standard; examples of relational database management systems; Oracle.

Module 4: Database Design Theory and Methodology: Functional dependencies and normalization for relational databases, relational database design algorithms and further dependencies.

Module 5: System Implementation Techniques: Query processing and optimization, transaction processing concepts, concurrency control techniques, database recovery techniques.

Module 6: Emerging Applications: Distributed databases and client/server models, XML Database (DTD, XML Schema), Query for XML Database, NoSQL.

**Text Book:**

1. *Fundamentals of Database Systems*, R. Elmasri, S. B. Navathe, Prentice Hall, New Delhi, 2007.

**Reference Books:**

1. *Database System Concepts*, A. Silberschatz, H. F. Korth, S. Sudharshan, Tata McGraw Hill, New Delhi, 2005.
2. *Introduction to Database Systems*, C. J. Date, Prentice Hall, New Delhi, 2004.

Tentative List of Lab Programs

Lab Program 1 involves designing an ER diagram for a real-world application such as a library or hospital, identifying entities, attributes, relationships, and cardinalities.

Lab Program 2 focuses on converting ER diagrams into Enhanced ER (EER) diagrams and then mapping them to relational schemas.

Lab Program 3 includes creating tables with constraints in SQL and performing basic operations like inserting, updating, deleting, and querying data using `SELECT` statements with `WHERE`, `GROUP BY`, and `ORDER BY` clauses.

Lab Program 4 requires students to write advanced SQL queries using joins (inner, outer), nested queries, and set operations to retrieve data from related tables.

Lab Program 5 introduces file organization and indexing concepts, where students simulate primary file organizations and create indexes to demonstrate single-level and multilevel indexing.

Lab Program 6 covers relational algebra and calculus, requiring students to perform operations such as selection, projection, and joins manually, along with writing relational calculus expressions.

Lab Program 7 focuses on database normalization, identifying functional dependencies, and applying 1NF, 2NF, 3NF, and BCNF to optimize table designs.

Lab Program 8 introduces transaction management using SQL commands like `BEGIN`, `COMMIT`, and `ROLLBACK`, and demonstrates concurrency control using locking and isolation levels.

Lab Program 9 allows students to analyze query performance using `EXPLAIN` plans in SQL databases, exploring different query strategies and indexing effects.

Lab Program 10 involves working with NoSQL databases such as MongoDB for document storage and creating XML documents with DTD or XML Schema and querying them using XPath or XQuery.

|             |                   |               |
|-------------|-------------------|---------------|
| Course Code | Course Name       | L – T – P : C |
| CS448       | Operating Systems | 3 – 0 – 2 : 4 |

**Objective:** This course provides undergraduate students with knowledge about contemporary operating system design and its relationship between memory, processor, file interface and processes.

**Learning Outcomes:** On successful completion of this course, students should be able to:

- Describe process management and concepts of threading, multitasking, IPC.
- Differentiation of various scheduling algorithms and identify the reasons of deadlock and their remedial measures in an operating system.
- Describe various memory management techniques, file system interfacing and disk scheduling algorithms.
- Conceptualize the components involved in designing a contemporary OS.



**Prerequisites:** None.

**Content:**

Module 1 - Introduction: Overview and history of Operating System, functions of an OS, device drivers, I/O interrupts, and system call interface.

Module 2 - Process Management: Process Abstraction, Process States and PCB, Threads, Process Scheduling; Real Time Scheduling; Classical Synchronization Problems; Semaphores; Monitors; Deadlocks, Deadlock Prevention, Avoidance and Detection.

Module 3 - Memory Management: Segmentation, Demand Paging, Hardware Support, Page Fault Handling; Page Replacement Algorithms; Shared Memory.

Module 4 - Storage Management: Disks Structure, Disk Scheduling, Disk Management, RAID Structure, File Organization and Mounting, File descriptor, Directory Structure and Implementation, I/O Systems.

**Text Books:**

1. Operating Systems: internals and design principles, William Stallings, Pearson Education.
2. Modern Operating Systems, Andrew Tanenbaum, Prentice Hall Publication.

**Reference Books:**

1. Operating System Concepts, A. Silberschatz, P. Galvin, & G. Gagne, John Wiley & Sons.
2. Operating System – A Concept based Approach, D.M. Dhamdhere, McGraw Hill Publication.

**Tentative List of Lab exercises**

Lab Exercise 1 introduces the basics of operating systems by exploring their evolution, functions, and the role of system calls and device drivers in hardware communication.

Lab Exercise 2 involves writing a simple program to demonstrate the creation and management of processes using system calls like `fork()`, `exec()`, and `wait()`.

Lab Exercise 3 focuses on implementing basic process scheduling algorithms such as First-Come-First-Serve (FCFS), Shortest Job First (SJF), and Round Robin.

Lab Exercise 4 addresses classical synchronization problems like Producer-Consumer and Reader-Writer using semaphores or mutex locks.

Lab Exercise 5 introduces thread creation and synchronization using POSIX threads (pthreads) or Java threads.

Lab Exercise 6 demonstrates deadlock detection and recovery strategies by simulating resource allocation and detecting circular wait conditions.

Lab Exercise 7 explores memory management by implementing page replacement algorithms like FIFO, LRU, and Optimal.

Lab Exercise 8 involves simulating demand paging and page fault handling with a basic virtual memory setup.

Lab Exercise 9 covers file system concepts by simulating file allocation methods such as contiguous, linked, and indexed allocation.

Lab Exercise 10 allows students to analyze disk scheduling algorithms such as FCFS, SSTF, SCAN, and C-SCAN, and understand the impact on seek time and performance.

|             |                   |               |
|-------------|-------------------|---------------|
| Course Code | Course Name       | L – T – P : C |
| HS442       | Technical Writing | 0 – 1 – 2 : 2 |

**Course Objectives:** This course equips IT students with essential communication skills, focusing on grammatical accuracy, technical vocabulary, and professional writing. It emphasizes effective report structuring, presentation skills for global teams, and verbal/non-verbal communication for meetings and interviews. Students will also gain hands-on experience with LaTeX, citation tools like Mendeley/Zotero, and understand plagiarism issues in technical writing and coding.

**Learning Outcomes:** By the end of this course, students will be able to:

- Communicate technical ideas clearly in writing and speech.
- Present IT topics effectively to varied audiences.
- Use professional verbal and non-verbal communication.
- Support content with visuals, examples, and code.
- Demonstrate confidence, clarity, and leadership.
- Avoid plagiarism and apply proper citation methods.
- Use LaTeX and follow IT-specific writing conventions.

**Prerequisite:** None

### Content

Module 1: Foundations of Technical and Professional Communication- Grammar and Writing Mechanics: Sentence structure, paragraph development, logical sequencing, and coherence. Audience-Centric Communication: Tailoring documentation for developers, end-users, and business stakeholders. Overcoming Communication Barriers: Addressing technical jargon, cross-cultural communication, and remote collaboration challenges.

Module 2: Technical and Scientific Writing- Technical Writing: Structure, style, and clarity in writing for software documentation, project reports, and research papers. Scientific and Technical Writing: Strategies for software documentation, white papers, and research papers. Referencing and Citation Styles: MLA, IEEE, and APA formats commonly used in technical research.

Module 3: Professional Communication in Industry- Professional Communication: Writing emails, bug reports, documentation for APIs, and system design proposals. Structuring Reports and Presentations: Logical sequencing, effective introductions, and strong conclusions. Confidence and Leadership in Communication: Speaking persuasively in professional settings, stand-up meetings, and stakeholder discussions.

Module 4: Visual and Data-Driven Communication- Use of Visual Aids and Infographics: Incorporating diagrams, charts, and flowcharts. Data Presentation and Analysis: Basics of univariate and multivariate analysis for technical reporting in IT. Technical Presentation Skills: Engaging an audience through clear articulation, demonstrations, and live coding sessions.

Module 5: Effective Presentation and Delivery Techniques - Delivery Techniques for IT Professionals: Presenting technical ideas clearly, handling Q& A sessions, and client interactions. Confidence and Leadership in Communication: Speaking persuasively in professional settings, stand-up meetings, and stakeholder discussions.

### **Text Book**

1. The Elements of Style – W. Strunk & E. B. White (New York: Macmillan, 1972).

### **Reference Books**

2. The Mayfield Handbook of Technical and Scientific Writing – L. Perelman (Mayfield Publishing Company, 1998).
3. The Science of Scientific Writing – G. D. Gopen & J. A. Swan (American Scientist, 78(6):550-558, Nov-Dec 1990).
4. Technical Writing: Process and Product – S. J. Gerson & S. M. Gerson (Pearson, 2017).
5. Handbook of Technical Writing – G. J. Alred, C. T. Brusaw, & W. E. Oliu (Bedford/St. Martin's, 2018).
6. Academic Writing: A Handbook for International Students – S. Bailey (Routledge, 2021).
7. They Say, I Say: The Moves That Matter in Academic Writing – G. Graff & C. Birkenstein (W. W. Norton & Company, 2021).
8. A Manual for Writers of Research Papers, Theses, and Dissertations – K. L. Turabian (University of Chicago Press, 2018).
9. How to Write and Present Technical Information – C. T. Brusaw (Macmillan, 1997).